

Formule de Taylor-Young.

Soit f une fonction de classe $\mathcal{C}^n$ sur un intervalle I . Alors f admet un $DL_n(a)$ en tout point a de I et on a

$$f(x) = f(a) + f'(a)(x-a) + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + o_{x \rightarrow a}((x-a)^n) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k + o_{x \rightarrow a}((x-a)^n)$$

$$\text{ou encore, } f(a+h) = f(a) + f'(a)h + \dots + \frac{f^{(n)}(a)}{n!}h^n + o_{h \rightarrow 0}(h^n) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}h^k + o_{h \rightarrow 0}(h^n).$$

Développements limités usuels en 0

$$\heartsuit \quad e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + o(x^n)$$

$$\heartsuit \quad \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$$

$$\heartsuit \quad \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

$$\text{th}x = x - \frac{x^3}{3} + \frac{2}{15}x^5 - \frac{17}{315}x^7 + o(x^8)$$

$$\heartsuit \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!} + o(x^{2n+1})$$

$$\heartsuit \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

$$\heartsuit \quad \tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \frac{17}{315}x^7 + o(x^8)$$

$$\heartsuit \quad (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}x^n + o(x^n)$$

$$\heartsuit \quad \frac{1}{1+x} = 1 - x + x^2 + \dots + (-1)^n x^n + o(x^n)$$

$$\heartsuit \quad \frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + o(x^n)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{1}{8}x^2 + \dots + (-1)^{n-1} \frac{1 \times 3 \times 5 \times \dots \times (2n-3)}{2^n n!} x^n + o(x^n)$$

$$\frac{1}{\sqrt{1+x}} = 1 - \frac{x}{2} + \frac{3}{8}x^2 + \dots + (-1)^n \frac{1 \times 3 \times 5 \times \dots \times (2n-1)}{2^n n!} x^n + o(x^n)$$

$$\heartsuit \quad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots + (-1)^{n-1} \frac{x^n}{n} + o(x^n)$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} + \dots + (-1)^n \frac{x^{2n+1}}{2n+1} + o(x^{2n+2})$$

$$\arcsin x = x + \frac{1}{2} \frac{x^3}{3} + \frac{3}{8} \frac{x^5}{5} + \dots + \frac{1 \times 3 \times 5 \times \dots \times (2n-1)}{2^n n!} \frac{x^{2n+1}}{2n+1} + o(x^{2n+2})$$